Date: 27 July 2022

BRAC UNIVERSITY

Department of Computer Science Engineering
MID TERM EXAMINATION, Summer 2022
CSE 350: Digital Electronics and Pulse Technique

Total Marks: 60 ($\mathbf{20 \times 3}$) **Time: 1 hour 30 minutes**

- *Total no of questions is **Four (4).** Answer any **Three (3)**.*
- *Figure in bracket [ ] next to each question indicates marks for that question.*
- *There is no need to draw circuits in your answer sheet.*

Name:	ID:	Section:

Question No. 1

For the **RTL** circuit in **Figure 1** the parameters are $\boldsymbol{\beta_F = 40, V_{CE}(Sat) = 0.2V, V_{OH} = 3\ V, V_{BE}(sat) = 0.8\ V, V_{BE}(active) = 0.7\ V}$ and $\boldsymbol{V_{OL} = 0.2\ V}$.

(a)	CO1	Briefly **explain** the operating mode of the transistor $\boldsymbol{T_1, T_2}$ and $\boldsymbol{T_3}$ when the input is logical high?	[6]
(b)	CO2	If the input is logical low and we connect **6 load** circuits to the driver circuit **calculate** the output voltage in V.	[8]
(c)	CO2	**Justify** (with the necessary calculation) the operating mode of the $\boldsymbol{T_2}$ transistor for the case of (b).	[6]

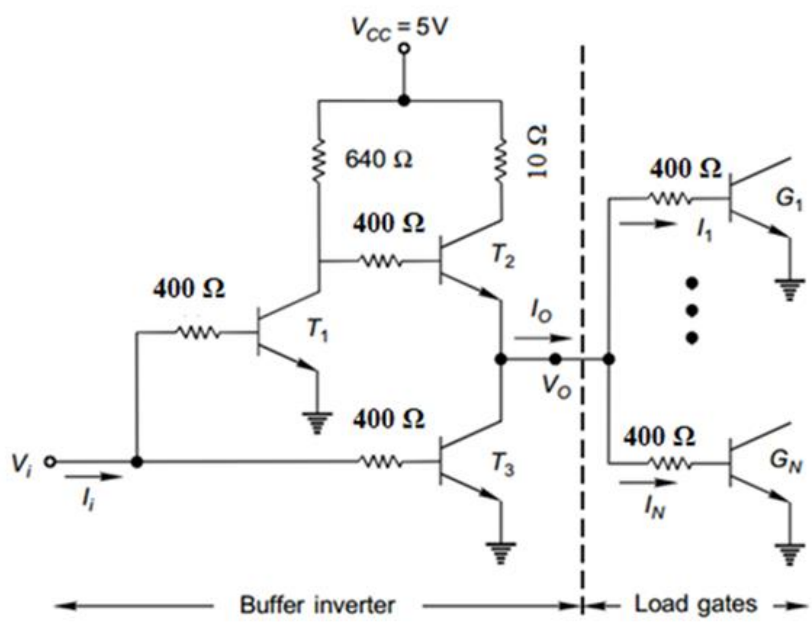

Figure 1

Question No. 2

The circuit in **Figure 2** has the following parameters: $\boldsymbol{\beta_F = 30, \beta_R = 0.1, V_{BE}(sat) = 0.8\ V,}$ $\boldsymbol{V_{BE}(active) = 0.7\ V}$ and $\boldsymbol{V_{CE}(Sat) = 0.1V}$.

(a)	CO2	**Determine** i_1, i_{B2}, i_2, i_{B0}, i_3 if all the inputs are high that means $\boldsymbol{v_x = v_y = 6V}$. [Assume no load connected]	[5]
(b)	CO2	**Identify** maximum power dissipation in $\boldsymbol{mW}$ [when all inputs are high]	[4]
(c)	CO2	1) **Find** the maximum fanout if any input of the **load= 0.1V**. 2) If both inputs of **load circuits= 0.1 V**, then what would be the maximum fanout? [Assume all inputs of the driver circuit are **high for above two cases**] Compare and comment on the above two cases to identify which case has better fanout and why.	[5+5+1]

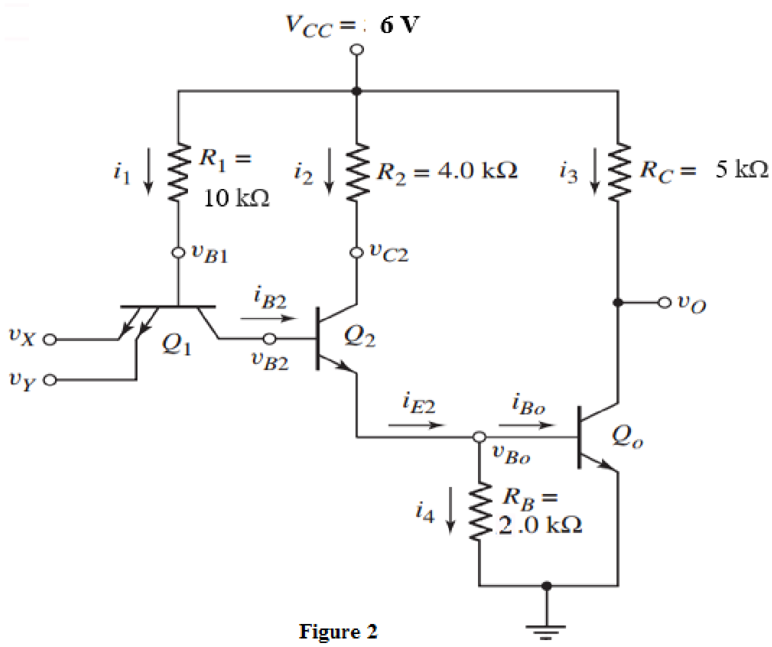

Figure 2

Question No. 3

In **Figure 3** the circuit has the following parameters: $\boldsymbol{\beta_F = 30, V_{BE}(Sat/Act) = 0.7V, V_{CE}(Sat) = 0.1V, V\gamma = 0.5V, V_D = 0.7V}$, Zener breakdown voltage, $\boldsymbol{V_{ZN} = 5.1V.}$

(a)	CO1	Briefly explain the advantages and disadvantages of **HTL** circuit over the modified **DTL** circuit.	[4]
(b)	CO2	When $\boldsymbol{V_A = 15V}$ and $\boldsymbol{V_B = 15V}$, determine the output voltage $\boldsymbol{v_o}$ and the operating modes of **T1, T2**. Repeat the process when $\boldsymbol{V_A = 0.1V}$ **and** $\boldsymbol{V_B = 15V}$ and determine $\boldsymbol{V_O}$ and the operating modes of **T1, T2.**	[6]
(c)	CO2	Considering the two input configurations mentioned in part (b), determine the noise margin of the **HTL** circuit.	[10]

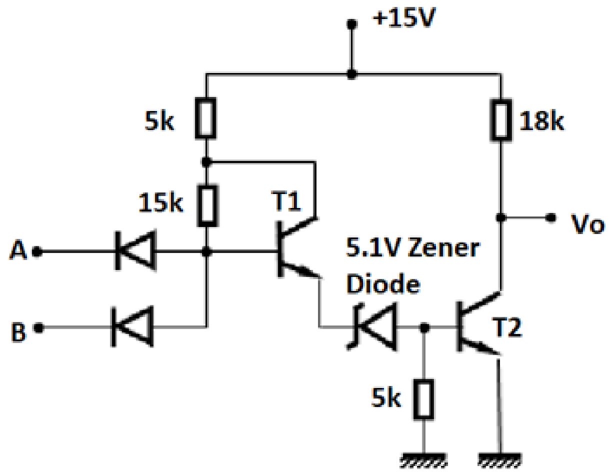

Figure 3

Question No. 4

For the DTL circuit in **Figure 4**, the following parameters are given:
$\boldsymbol{V_{BE}(Sat) = 0.8V, V_{CE}(Sat) = 0.2V, V_{\gamma}(transistor) = 0.5V, V_{\gamma}(diode) = 0.6V, \beta_F = 40}$.

(a)	CO2	Assume inputs **A, B** and **C** have logical low values, and no load is connected to the driver circuit. Use nodal analysis to find the voltage at point **P** and the current $\boldsymbol{I_L}$.	[7]
(b)	CO1	Assume all the inputs have logical high values, and no load is connected to the driver circuit. Explain and verify the operating mode of transistor **Q**.	[8]
(c)	CO2	For the case mentioned in (b), calculate the power dissipation in $\boldsymbol{mW}$.	[5]

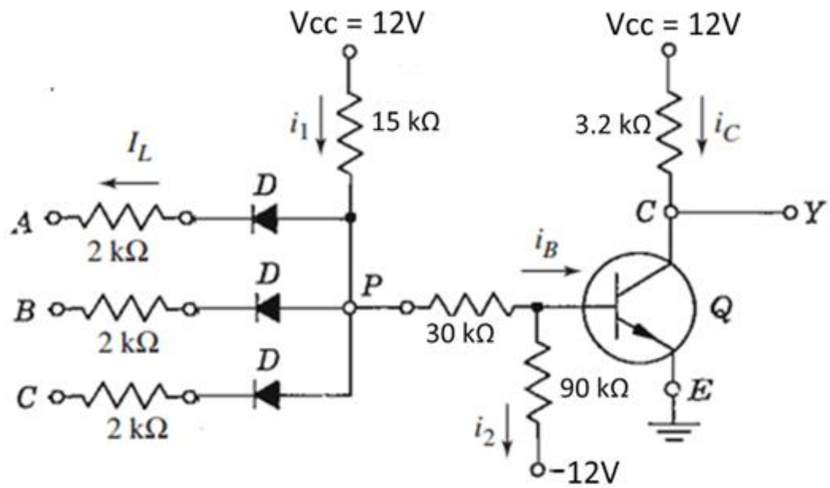

Figure 4